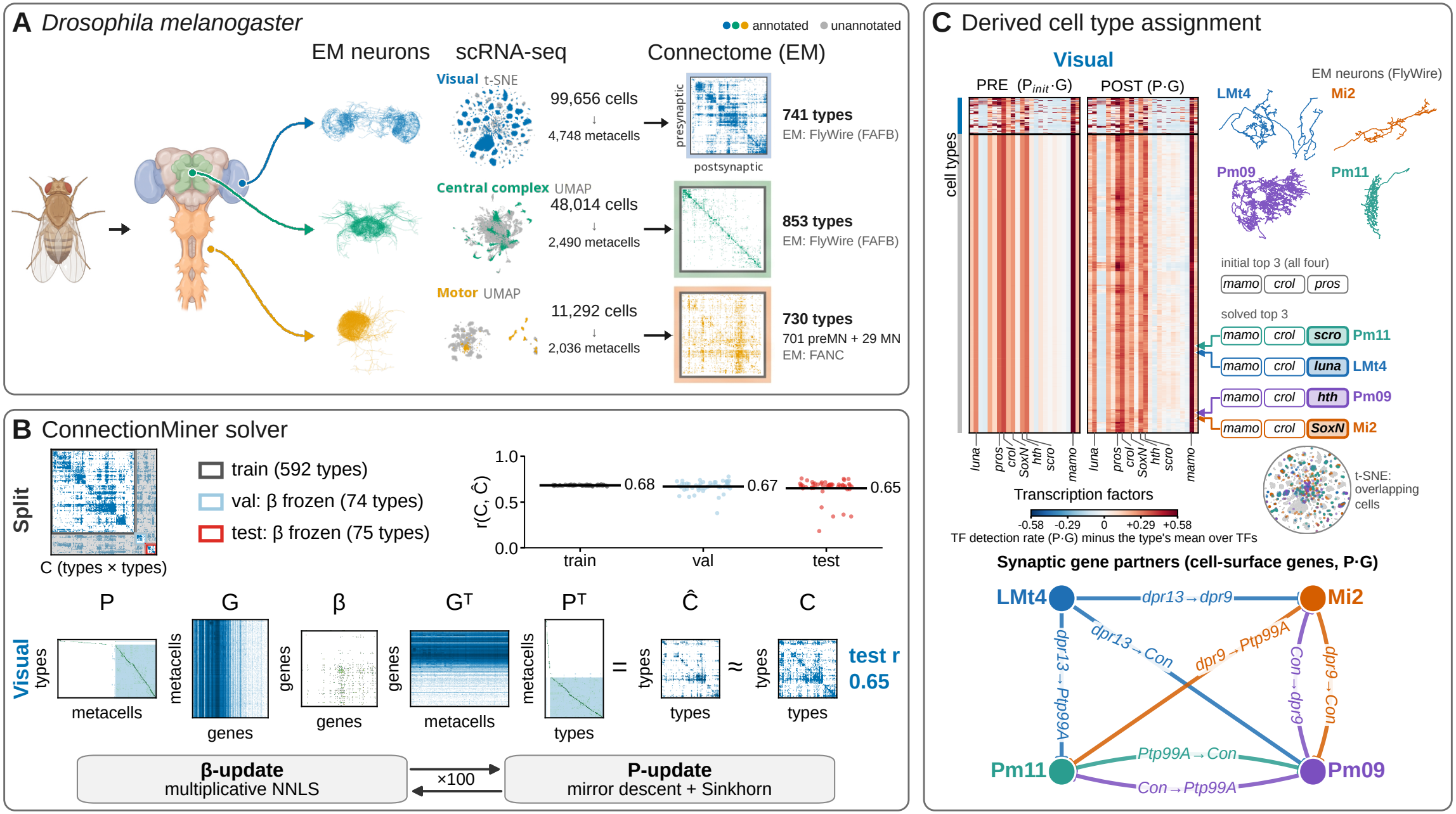

A Drosophila melanogaster
annotated
unannotated
EM neurons
scRNA-seq
Connectome (EM)
Visual t-SNE
99,656 cells
4,748 metacells
presynaptic
postsynaptic
741 types
EM: FlyWire (FAFB)
Central complex UMAP
48,014 cells
2,490 metacells
853 types
EM: FlyWire (FAFB)
Motor UMAP
11,292 cells
2,036 metacells
730 types
701 preMN + 29 MN
EM: FANC
B ConnectionMiner solver
Split
C (types × types)
train (592 types)
val: β frozen (74 types)
test: β frozen (75 types)
r(C, Ĉ)
1.0
0.5
0.0
0.68
0.67
0.65
train
val
test
P
G
β
G^T
P^T
Ĉ
C
Visual
types
metacells
genes
test r
0.65
β-update
multiplicative NNLS
×100
P-update
mirror descent + Sinkhorn
C Derived cell type assignment
Visual
PRE (P_init·G)
POST (P·G)
cell types
EM neurons (FlyWire)
LMt4
Mi2
Pm09
Pm11
initial top 3 (all four)
mamo
crol
pros
solved top 3
scro
luna
hth
SoxN
Transcription factors
-0.58 -0.29 0 +0.29 +0.58
TF detection rate (P·G) minus the type's mean over TFs
t-SNE: overlapping cells
Synaptic gene partners (cell-surface genes, P·G)
dpr13→dpr9
dpr13→Con
dpr9→Ptp99A
dpr13→Ptp99A
Con→dpr9
dpr9→Con
Ptp99A→Con
Con→Ptp99A